Status	Finished
Started	Sunday, 18 October 2026, 7:31 PM
Completed	Sunday, 18 October 2026, 7:52 PM
Duration	21 mins 11 secs
Grade	**11.00** out of 15.00 (**73.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The first element is returned
c. The search is considered unsuccessful ✓
d. The last element is returned

Question **2**

Incorrect

Mark 0.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Bubble search ✗
c. Merge search
d. Linear search

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is the last element in the array
c. When the item is somewhere in the middle of the array ✓
d. When the item is not the array at all

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the left sublist ✓
c. In the right sublist
d. In the entire list

Question **5**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 90 and 99
c. 89 and 94
d. 94 and 99 ✗

Question **6**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Hashing
c. Linear search ✓
d. Jump search

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The search stops
c. The list is sorted
d. The search continues from the beginning

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Interpolation search
c. Linear search ✓
d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It works on unsorted lists
d. It compares elements sequentially

Question **10**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the middle element in the list
c. Sort the list ✗
d. Compare the target element with the first element in the list

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. log n
- d. n/2

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O(log n)
- c. O(n log n)
- d. O👎 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search works on unsorted lists
- b. Binary search can find multiple instances of the target element
- c. Binary search has a lower time complexity on large, sorted lists ✓
- d. Binary search does not require dividing the list

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It does not work with negative numbers